

Stock Market Forecasting

A Comparative Study of Predictive Validity and Overfitting in Stock Market Forecasting Models

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Abstract

Can a machine-learning model really learn to predict tomorrow's stock return, or can apparently forecasts simply reflect patterns that happened to occur in the past? This dissertation investigates this question using daily returns of Apple Inc. (AAPL), comparing a simple Linear Regression model with a more flexible Long Short-Term Memory (LSTM) network. The study is motivated by the difficulty of distinguishing genuine predictive information from noise and overfitting in financial markets, where historical patterns can appear meaningful even when they have little value for future observations.

Rather than judging the models only by their forecasting accuracy, the study evaluates them against a simple mean-return benchmark and examines whether their performance remains convincing when tested on unseen data. The same modelling framework is also applied to a synthetic random-return series containing no intentionally introduced predictive structure. This provides a useful comparison for asking whether apparent forecasting ability can also arise when there is no genuine signal to learn.

The LSTM achieves the lowest forecasting error among the fitted models, but its improvement over simpler alternatives is small and does not provide convincing evidence of a meaningful predictive advantage. The simple benchmark performs surprisingly competitively, particularly when the direction of the next return is considered. Models trained on the random data do not produce strong predictive performance, supporting the expectation that flexible models do not automatically discover useful patterns in pure noise.

Overall, the study does not find strong evidence that historical AAPL price and volume information provides a reliable source of next-day return predictability. More importantly, the findings demonstrate why a low forecasting error alone should not be interpreted as evidence of genuine predictive ability. Comparing against simple benchmarks, testing on unseen periods and examining how models behave under a no-signal benchmark provide a more cautious way of evaluating financial forecasting models.

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1. Introduction

1.1 Research context

Financial forecasting is attractive precisely because it is difficult. Even a small and repeatable improvement in predicting future returns could be valuable when multiplied by many data points. At the same time, this context presents several features that make false discoveries very easy such as the signal-to-noise ratio is low, return distributions are heavy-tailed, volatility changes through time, modelling choices are numerous, and repeated experimentation can unintentionally tune a model to historical accidents. The key issue is then not just whether the model forecasts, but whether its forecasting ability generalises to new data.

The Efficient Market Hypothesis (EMH) is a natural theoretical benchmark which states that all assets are priced to show any new or recent information. This does not give any window to capture excess returns. However, traders who can exploit this time gap within which the market is inefficient, can earn extra returns. It can be said that trading is the way in which the new information is incorporated in the asset prices. The speed with which information is adjusted is actually the time taken for the trade to get executed. There are many people who use for researching and investing in stocks in various markets. Features such as give access to a broad array of global stocks to investors. The investors in the market can behave either rationally or normally. In case of abnormal information, the investors will behave abnormally, and it is normal behaviour, but behaving as everyone else does is also normal behaviour. The stock price reflects all the information that is relevant to the investors. It also means that the investors cannot take advantage of the market because they have to behave according to the information in the market.

According to its weak form, prices are supposed to contain all the information embedded in past prices, which means that past price behaviour cannot provide any easily exploitable forecasting rule (Fama 1970). This does not imply that all conditional forecasts have to be identically zero or that markets are efficient at any moment of time. What this implies is that systematic, risk-adjusted and economically meaningful forecast ability should be difficult to extract from public price history alone. There have been some empirical studies on mean-reverting behaviour at longer horizons and other violations of the random walk hypothesis (Fama and French 1988), while noise trading even allows prices to deviate from their fundamentals without making it easy to exploit them (Black 1986).

Modern machine learning makes this problem more interesting. Flexible algorithms can approximate nonlinear mappings and interactions that a simple linear model cannot represent. Cross-sectional studies have shown that neural networks and tree-based approaches can add predictive value when large panels of firm characteristics are available (Gu et al. 2020), and LSTM architectures have produced strong results in some equity-direction settings (Fischer and Krauss 2018). However, greater flexibility also expands the set of patterns a model can fit. A sufficiently complex model may fit noise in a finite sample and report an impressive historical score while failing when market conditions change.

Predictive validity and overfitting are thus both aspects of the same empirical investigation in this thesis. Rather than pursuing the best possible in-sample fit, the comparison is made between the naïve approach, an interpretable linear model, and a recurrent neural network using a hold-out set chronologically ordered. Furthermore, the performance of these models is compared between actual returns from AAPL and a return series simulated in the absence of any signal. The study deliberately reports negative and near-zero results because, in a domain

affected by data snooping, a careful failure to find strong predictability can be more informative than an unvalidated claim of success.

1.2 Motivation and research gap

The increasing application of machine learning in finance raises a crucial methodological issue. When a more flexible model makes better forecasts in history, it is natural to take the increase in forecast performance as a sign that the model has found information inaccessible to simpler procedures. But in finance, the same outcome may be obtained because the model has learned to predict some historical sample.

Thus, much of the appeal of machine learning in finance centers on the question of whether such models may improve predictions. But a second equally relevant issue is whether a given improvement will hold in the face of truly unseen data or of data where no prediction process was built in. This last issue becomes especially relevant when the signal is small compared to daily market fluctuations.

The current research fills this void by framing the issue of forecasting performance in terms of validation and not merely model selection. Linear Regression offers a simple and interpretable baseline model, and the LSTM offers the more complex sequence-based approach. The models will be compared on the basis of their performance relative to a naive benchmark and a randomly generated experiment involving returns. In this way, the current research is able to ask not only whether one model works better than another, but whether this difference offers compelling evidence of generalizable information.

This is a critical point since lack of predictive power is not an irrelevant finding in a very noisy environment. If one can show that even the seemingly superior model does not offer compelling evidence of predictive power, then this is useful information on the limits of forecasting.

1.3 Problem Statement

The issue of backtest overfitting arises from model selection and parameter optimisation being based on specific characteristics of the historical data set. Data snooping was defined by White (2000) as the problem of assessing several rules using the same data set where the best-performing rule may appear to be superior just because the data was used multiple times. Bailey et al. (2017) advanced methodology for estimating the probability of backtest overfitting and showed that the usual method of hold-out is flawed when many strategies are tested. In general, Harvey et al. (2016) explain why standard significance levels are inappropriate when a great number of financial predictors or models are evaluated.

This problem is especially true for predicting stock returns since the past financial data have plenty of noises and weak signal strength. The model that performs well on prediction within the sample set may be discovering patterns from history that would not work on other samples rather than learning the relationship between inputs and outputs. Higher complexity of the model increases this problem especially in case of flexible models like LSTM network.

Accordingly, the performance of a model cannot be considered based simply on minimum prediction error. Firstly, the performance of the prediction should be measured with respect to out-of-sample data relative to some naive benchmark. Secondly, residual diagnostics can be carried out to see if there is any systematic structure left in the residual after making predictions. Thirdly, the difference in predictive performance among different competing models should be examined statistically and through effect size, rather than simply by the small difference in error measure values. Fourthly, synthetic random data can be another benchmark to explore

whether a modelling process generates apparent prediction even though no predictive method is built into the process.

These facts form the basis for the need for this research, which tests the performance of Linear Regression and LSTM against actual returns data from AAPL along with artificial random return data. This research is therefore aimed at not only establishing the superiority of one of the models in terms of lower prediction errors but also determining whether the predictive power is real or simply due to model complexity.

1.4 Aim and research questions

The purpose is to determine if statistical and machine learning models produce consistent short-term forecasts for daily Apple (AAPL) stock returns and if their perceived ability to do so is more consistent with actual predictability or mere overfitting. The study compares model performance using real financial data and a synthetic random-return benchmark, with particular emphasis on out-of-sample predictive accuracy and the difference between in-sample and out-of-sample performance. The analysis addresses three research questions:

- **RQ1:** To what extent can statistical and machine-learning models accurately predict stock returns using historical market information?
- **RQ2:** How does the difference between in-sample and out-of-sample performance inform the diagnosis of overfitting?
- **RQ3:** Can predictive performance observe on real financial data be distinguished from patterns produced by simulated random data?

1.5 Contributions

The dissertation makes four modest but useful contributions. First, it provides a comparative evaluation of models with different levels of complexity, namely Linear Regression and LSTM, using a common chronological training and testing framework and a consistent set of predictive performance metrics. This ensures that model performance is assessed using out-of-sample observations rather than relying solely on in-sample fit.

Second, the study complements conventional predictive-performance measures with a range of diagnostic and statistical analyses. In addition to MSE, RMSE, MAE and R^2 , the analysis considers directional accuracy, residual behaviour and statistical tests of paired prediction errors. This provides a broader assessment of whether differences in predictive performance are meaningful rather than simply selecting the model with the smallest numerical error.

Third, the dissertation introduces an explicit synthetic random-return benchmark. The same modelling framework is applied to data generated without an intentionally introduced serial predictive mechanism, allowing the performance obtained from real AAPL returns to be compared with performance that can arise under a random-data setting. This provides an additional empirical benchmark for assessing whether apparent predictive ability is associated with genuine structure in the financial data or may arise from the modelling process itself.

Fourth, the study conducts a transparent post-implementation audit of the research design. Rather than presenting the executed analysis as identical to the original proposal, discrepancies between the proposed methodology and the implemented notebook are explicitly identified as

potential threats to validity. These discrepancies are used to define specific methodological improvements and requirements for future research.

2. Background and Related Work

2.1 Market efficiency and the meaning of predictability

The EMH is often reduced to the statement that prices are unpredictable. A better interpretation is information-based, where a market is said to be efficient with respect to an information set if the price already reflects the information and abnormal gains can no longer be obtained using this information (Fama 1970). When we consider the weak form, the relevant information set consists of past prices and returns. The current dissertation then questions whether lagged returns and technical-like rolling features have any additional information worth using for one-day ahead predictions.

There are no requirements regarding normality or independence of returns in the weak-form criterion. Financial returns might exhibit conditional heteroscedasticity and risk that varies with time and small autocorrelation without providing any stable trading opportunity. For instance, Fama & French (1988) provide empirical evidence of mean reversion which grows in significance when the horizon increases. The reason why it is relevant to the present model is that a day-to-day forecast is a particularly tough task: a phenomenon that looks quite reasonable on the multi-year horizon might have no practical application on a day-to-day horizon. Moreover, noise plays an important economic role according to Black (1986). Noise may distort prices but make the process of inference complicated for researchers as well.

As such, a rather conservative approach to defining predictability success will be employed hereinafter. Predictability validity cannot be established just by virtue of having the numerically lower MSE compared to the other model. More convincing proof would entail a better performance out of sample compared to an appropriate naive benchmark along with informative residual diagnostic testing and statistical evaluation of differences in the forecasting error. The comparison with the random-return synthetic benchmark provides an additional criterion to check if the observed predictability success can be obtained in the case where there was no attempt at introducing any serial predictability at all.

However, economic relevance would imply an extra stage of analysis considering transaction costs, risk, and possibly portfolio performance.

2.2 Statistical features of financial returns

There are significant differences between daily returns of the asset price and the normal assumption of errors being independent and normally distributed in many basic regression scenarios. For instance, the distribution of financial return is characterized by fat tails, where the probability of large observations is greater than what is observed in the case of a Gaussian distribution, as well as volatility clustering, whereby there are periods of high volatility that are followed by high volatility periods. The ARCH model, which was proposed by Engle in 1982, was used to capture the conditional variance, thus providing a useful platform in examining the volatility of financial time series. These aspects are important in modelling and evaluation in the sense that a model can minimize average squared error by performing better in stable periods but poorly in volatile periods of the markets.

The other crucial aspect of financial time series data is stationarity. While stock price levels are usually non-stationary, returns tend to be stationary since the transformation eliminates most of the price trend. In this case, the current research applies a transformation to the closing price data for AAPL to calculate daily percentage returns and then applies the Augmented Dickey-Fuller (ADF) and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) stationarity tests. This decision is justified by the fact that the null hypotheses for these two tests are complementary – while the ADF test checks for the presence of a unit root, the KPSS test checks for stationarity.

On the other hand, stationarity of the conditional mean does not mean the constant variance or lack of dependence between terms. Thus, the exploratory analysis will focus on ACFs and PACFs, along with the Ljung-Box tests performed on returns and squared returns. The ARCH-LM test will also be utilized to determine whether the process of conditional heteroscedasticity exists. This point is crucial: dependence in raw returns relates to the predictability of the conditional mean, but dependence in squared returns may be associated with the presence of volatility clustering, despite the conditional mean being hard to predict.

The distributional properties of the returns are tested via the calculation of skewness and kurtosis, the Jarque-Bera test for normality, histograms, box plots and the Q-Q plot. These measures allow us to get information about asymmetry, tail behaviour and deviation from normality. The mentioned properties play a crucial role in financial forecasting, since the extreme values can have considerable impact on the squared error-based measures.

Another point is related to the type of measures for assessing forecast quality. The Mean Squared Error (MSE) puts considerably more emphasis on the large forecasting errors as they are squared. This method can be especially helpful in cases where we believe large errors are of great importance, but it makes MSE sensitive to the presence of extreme values in heavy-tailed financial data. The Mean Absolute Error (MAE) and the Root Mean Squared Error (RMSE) give an alternative to the MSE. While the first is a linear measure with equal weights assigned to each error value, the latter represents a squared-error measure in the returns scale.

The provision of both complementary statistics and evaluation measures decreases the risk of evaluating the performance of the model on the basis of one particular statistic. In relation to the current dissertation research, this is especially relevant since the purpose of the research is not limited to finding the best-performing model in terms of numerical prediction accuracy, but to checking its predictive ability.

2.3 Linear models as disciplined baselines

Linear regression estimates the conditional mean of the target variable as a linear combination of the available predictors. Its primary value in this dissertation is not the assumption that financial markets are inherently linear, but its role as a low-complexity and interpretable benchmark. The model provides directly interpretable coefficients, allowing the contribution and direction of individual predictors to be examined. In the present study, the predictors comprise lagged returns, moving-average features, rolling volatility and changes in trading volume. If these historical and technical-style features contain stable linear information about the next-day return, Linear Regression should capture at least part of that relationship.

A linear regression model predicts a conditional mean by forming a linear function of the predictors. What is valuable about linear models here is that financial markets need not be linear, but the linear regression produces a simple baseline model where the coefficients can be checked. If the past returns have stable linear signals that allow for predicting the future, the

linear regression will capture at least some of them. On the other hand, if the highly complex LSTM does not beat a well-specified linear model much, there is little to be gained from complexity.

The bias-variance trade-off is crucial. While simple models can suffer from underfitting nonlinear relations, their simplicity makes them more generalisable in noisy data (Hastie et al. 2009). A complex model reduces approximation bias but increases variance of the estimates. When dealing with financial data where the target signal is weak in comparison with daily noise, the trade-off matters much. Thus, any predictive model should favour additional complexity only if it produces substantial and consistent improvements.

2.4 LSTM networks for financial sequences

Long Short-Term Memory (LSTM) networks are a type of recurrent neural network which attempts to address temporal dependencies and the problem of vanishing gradients present in conventional recurrent networks (Hochreiter and Schmidhuber, 1997). The gating mechanism inherent to LSTM networks makes them useful in dealing with sequential data, wherein the influence of recent observations could be influenced by time.

In this thesis, the LSTM model will serve as the benchmark model against the Linear Regression model. The model uses sequence of historical observations to predict the next day's return of AAPL. This is done to investigate if modelling the non-linear relationships would yield better results than the linear one.

However, increased flexibility of the model may lead to the problem of overfitting. For that reason, the performance of the LSTM is evaluated through out-of-sample error, directional accuracy, residual analysis, and comparison with the simulated random data. The LSTM is therefore not considered to be better just because it is more complicated.

LSTM Hyperparameters and Their Rationale

The LSTM hyperparameters were selected to provide sufficient model capacity to capture short-term temporal and nonlinear relationships while avoiding an unnecessarily complex architecture. The choices were also intended to support the dissertation's focus on generalisation and overfitting rather than maximising in-sample performance.

Sequence length: The 20-day sequence length is particularly important for this experiment. Rather than treating each observation independently, the LSTM receives a sequence containing the previous 20 observations. This gives the model an opportunity to identify temporal relationships between recent returns, lagged variables, moving averages and volatility. A longer sequence could potentially capture more distant dependencies, but it would also increase the amount of historical information presented to the model and potentially increase model complexity. A 20-observation window therefore provides a practical short-term forecasting horizon consistent with the study's focus on one-day-ahead prediction.

LSTM layers: The network uses 64 units in the first LSTM layer and 32 in the second. The first layer provides enough capacity to extract temporal patterns from the input sequences, while the second layer produces a more compact representation of those patterns. The decreasing structure from 64 to 32 units also avoids constructing an unnecessarily large network. This is relevant because the dissertation is specifically investigating whether apparent predictive performance is genuine or caused by overfitting.

Dropout: A 20% dropout rate was included after each LSTM layer. During training, dropout randomly removes a proportion of connections, preventing the network from becoming excessively dependent on particular internal representations. This is particularly relevant for financial forecasting because daily stock returns contain considerable noise and weak signal. Without regularisation, a flexible LSTM could potentially learn patterns that are specific to the training sample rather than patterns that generalise to unseen observations.

Adam optimiser: The Adam optimiser was selected because it adaptively adjusts the learning rate for different parameters and is computationally efficient for neural-network optimisation (Kingma and Ba, 2015). This allows the LSTM weights to be updated efficiently during training without requiring a manually specified learning-rate schedule.

MSE loss: The model minimises Mean Squared Error (MSE) is given by

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

MSE was selected because the objective is to predict a continuous numerical return. It is also directly comparable with the MSE reported in the final model evaluation. An important characteristic of MSE is that larger errors receive greater weight because the residual is squared. This makes the model particularly sensitive to large forecasting mistakes, which can be relevant when evaluating financial-return forecasts.

Early stopping: The model was trained for a maximum of 100 epochs, but early stopping with a patience of 10 epochs was used. Training therefore stops when validation loss does not improve for 10 consecutive epochs. This is an important methodological choice for the dissertation because a model that continues training after its validation performance has stopped improving may begin to fit sample-specific noise. Early stopping therefore acts as a form of regularisation and supports the study's analysis of overfitting.

2.5 Backtest overfitting, data snooping and multiple testing

Backtest overfitting arises when the model selection process or the process of optimizing parameters becomes excessively tuned to the historical data set. According to White (2000), data snooping is the process whereby different rules are assessed using the same data set, where in this case, the best rule turns out to be the one that takes advantage of idiosyncrasies within the data set rather than predictive power.

This problem arises especially in financial machine learning due to the possibility of varying several factors such as features, architectures, parameters, time horizons, and criteria used to evaluate models. By consistently picking the best performing model, there is a higher likelihood of selecting a spurious model due to chance alone. Harvey et al. (2016) point out that multiple testing arises in finance and require greater evidence for more alternatives being considered.

The current research paper tries to mitigate such problems through application of a fixed chronological train/test split, comparison with a naïve benchmark, and simulation of random return data set. The difference between models is estimated using residual analysis and paired test methods rather than simply based on the minimal forecasting error value. Such approach

will not help to eliminate overfitting, but will add extra tools for checking whether single out-of-sample forecasting improvement can be treated as true predictability.

2.6 Forecast evaluation and temporal validation

For financial forecasting, observations need to be divided based on their temporal order, to prevent any use of future data in developing the model. Thus, this research employs a temporal train/test split, where earlier observations will serve as training data while later observations are kept aside for test purposes. MinMax Scaler is trained on training data and then used on test data, minimizing the chances of information leakage.

However, a single chronological split has a limitation. It provides evidence from only one out-of-sample period and may therefore be sensitive to the market conditions at that particular point in time. Rolling-origin or expanding-window validation could provide a more robust assessment by evaluating the models across multiple periods. This is identified as an area for future work.

Finally, the study applies statistical comparison of model errors. The conducted analysis employs paired t-test and Wilcoxon signed-rank tests for absolute forecasting errors of Linear Regression and LSTM models. This gives additional information whether the found difference in forecasting errors is systematic. To conduct the forecasting-oriented analysis, the authors could use Diebold-Mariano test (Diebold and Mariano, 1995).

Overall, the validation strategy is designed to provide a minimum valid temporal evaluation while acknowledging that a single train-test split cannot establish that forecasting performance is stable across different market regimes.

3. Data and Exploratory Analysis

3.1 Data Source and Dataset Selection

Daily market data for Apple Inc. (AAPL) were obtained through the yfinance Python library from the Yahoo Finance data repository for the period 1 January 2015 to 1 January 2025. The initial download contained 2,516 daily observations covering Open, High, Low, Close and Volume (OHLCV) variables. Two early rows were subsequently removed during data cleaning. Following feature construction and the removal of observations made incomplete by lagged and rolling-window calculations, the final modelling dataset contained 2,493 observations, covering 4 February 2015 to 30 December 2024. The prediction target was defined as the next-day percentage return.

The first reason why AAPL was chosen for the empirical case study is both practical and methodological in nature. Firstly, a ten-year daily history of the firm provides a sufficient amount of data points needed to train machine-learning models as well as to evaluate model performance on the out-of-sample dataset that is temporally separated from training. The ten-year timeframe used for the analysis will also contain significantly different periods of time with respect to volatility and general market conditions, thus providing the possibility to examine the behaviour of the model under different market conditions rather than using a short time frame.

Another important reason why the firm is chosen as an object of study is related to the fact that it is one of the largest firms by capitalization and liquidity on NASDAQ. Thus, AAPL represents an appropriate case to test the predictability of short-term returns since this stock is actively traded and has a significant history of observed prices and trading volume. Nevertheless, such selection restricts the ability of generalizing the obtained results as the relationships observed for AAPL can be true only for the firm and might not work for other securities. Seven explanatory variables were created using AAPL return data which are Lag1, Lag2, Lag5, MA5, MA20, Volatility20, and VolumeChange. These correspond to recent return dynamics, short and medium-term trends, volatility and trading volume fluctuations respectively. In this case, the prediction target is tomorrow's return. Therefore, the exercise is a one-step ahead prediction problem.

In order to find out if any observed predictive performance is due to actual predictability or simply due to model fitting to noise, a random returns benchmark is employed. Both AAPL and the random returns will be tested using the same predictive pipeline. Predictive performance for AAPL but not for the random returns would suggest actual predictability. The dataset includes a long time series of daily observations, liquid market prices, suitable financial variables and comparison with random returns. It is, however, restricted to one security and one forecasting horizon only. Furthermore, the random returns used do not incorporate some of the characteristics of real financial time series like volatility clustering.

3.2 Return and feature construction

Daily simple returns are defined as

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

where P_t is the closing price at time t . The prediction target is the next-day return. Seven predictor variables are used: Lag1, Lag2, Lag5, MA5, MA20, Volatility20 and VolumeChange. Lagged returns capture recent short-term dependence, moving averages represent recent price trends, rolling volatility measures changing market risk, and volume change reflects changes in trading activity.

These features were selected because they provide economically interpretable measures of momentum, trend, volatility and market activity while keeping the model relatively simple. Their predictive usefulness is treated as an empirical question rather than assumed in advance. The predictors are scaled using a MinMaxScaler fitted only on the training data to prevent information leakage before applying the Linear Regression and LSTM models.

3.3 Descriptive statistics

The average daily return is positive but small relative to its standard deviation. The near-zero skewness indicates approximately symmetric central behaviour, yet excess kurtosis above five demonstrates much heavier tails than a Gaussian distribution. The symmetric distribution can still contain far more extreme observations than a normal model expects.

Statistic	Value
Observations	2,493
Mean	0.001065
Median	0.000989
Standard deviation	0.017885
Minimum	-0.128647
Maximum	0.119808
Skewness	-0.011088
Excess kurtosis	5.405624
Pearson kurtosis	8.405624

Table 3.1: Descriptive statistics for AAPL daily returns in the modelling dataset.

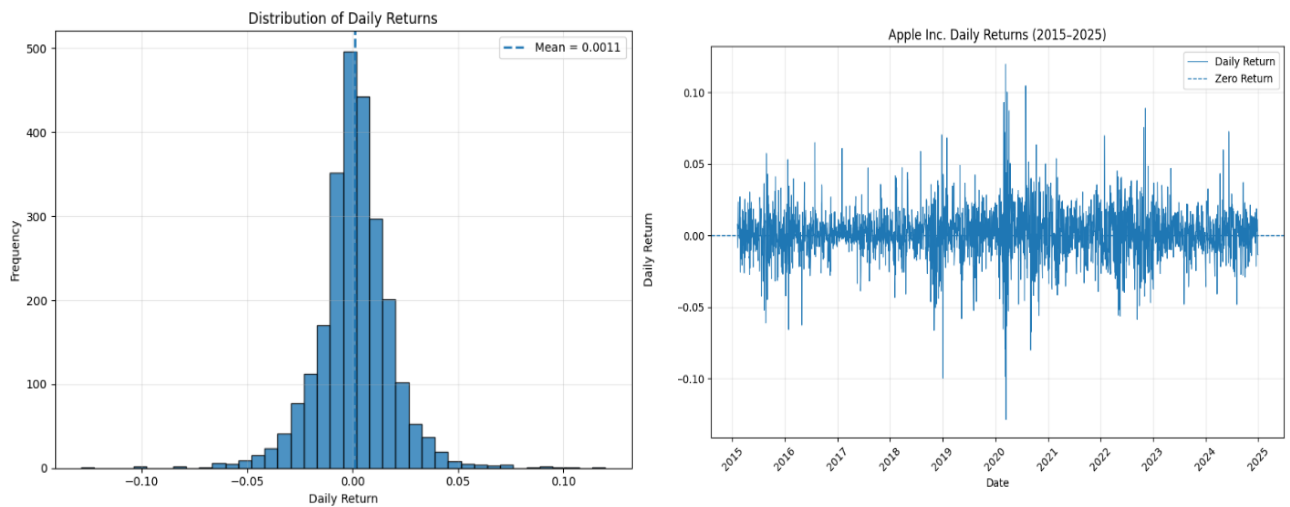


Figure 3.1: AAPL adjusted closing-price series and daily return series graphs. The price level trends strongly over the decade, whereas returns fluctuate around a small mean with time-varying dispersion.

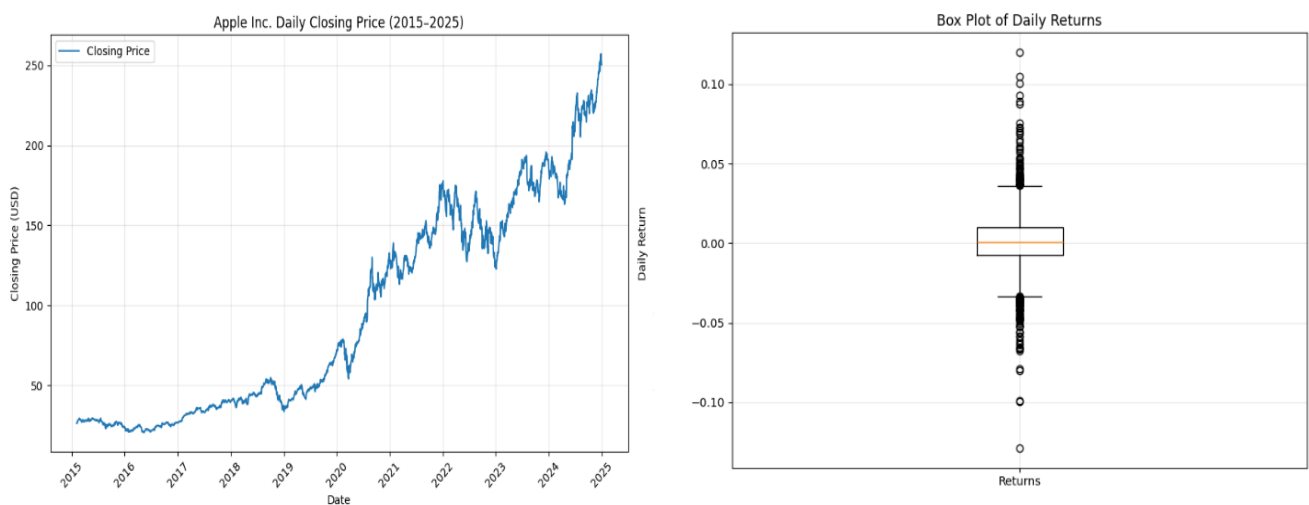


Figure 3.2: Distribution and box plot of daily AAPL returns. The concentration near zero co-exists with numerous large positive and negative observations.

Figure 3.1 visually separates two properties that motivate the modelling design. The price level is strongly non-stationary, while returns fluctuate around a much more stable location. Volatility is heterogeneous, periods around market stress show visibly larger return amplitudes than quieter intervals. This makes a simple train-test error gap difficult to interpret without considering target scale.

The Jarque-Bera statistic is 3020.51 with a p-value effectively zero at printed precision, strongly rejecting normality. Figure 3.2 shows the practical reason: most returns are near zero but the tails contain many large moves. This finding cautions against interpreting MSE differences through a Gaussian-error lens alone and motivates the complementary use of MAE and a non-parametric Wilcoxon comparison.

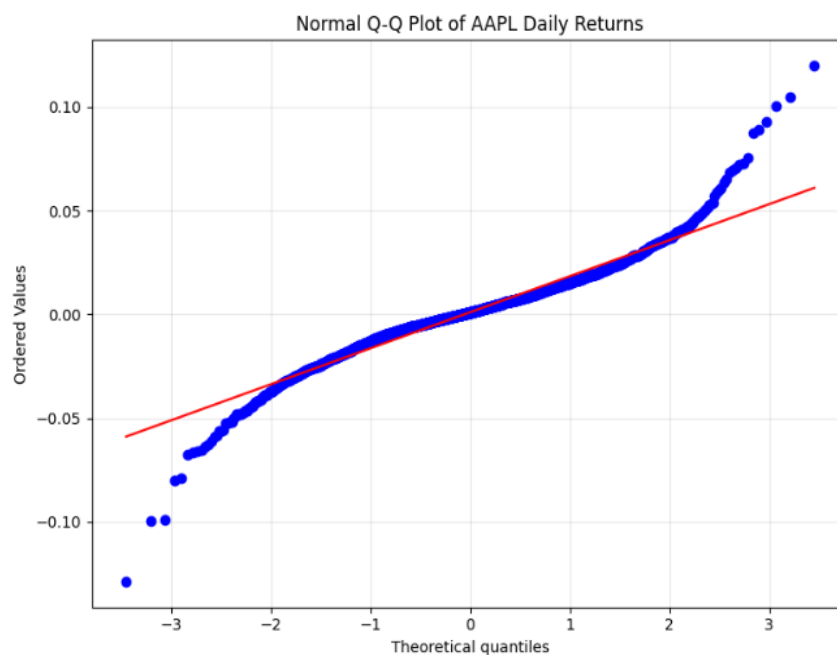


Figure 3.3: Normal Q–Q Plot of AAPL Daily Returns

The Normal Q–Q plot shows substantial deviations from the straight reference line, particularly in both tails. The extreme observations fall well above and below the line, indicating that AAPL daily returns have heavier tails than would be expected under a normal distribution. This is consistent with the excess kurtosis of 5.41 observed in the descriptive statistics and suggests the presence of extreme return events. The central portion of the distribution is relatively close to the reference line, but the strong tail deviations indicate that the daily returns do not follow a normal distribution.

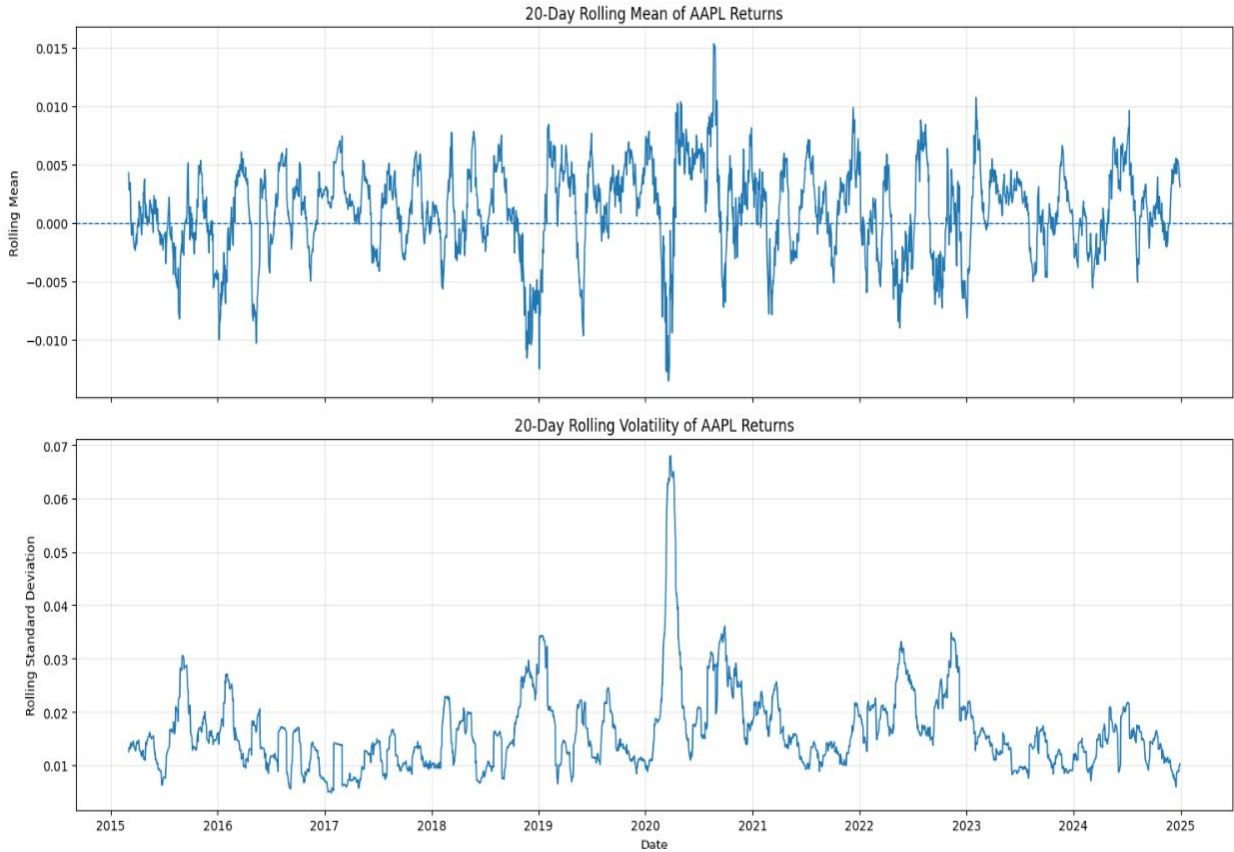


Figure 3.4: 20-Day Rolling Mean and Rolling Volatility of AAPL Returns

Figure 3.4 shows that the 20-day rolling mean of AAPL returns fluctuates around zero, while rolling volatility varies considerably over time. The pronounced volatility spikes, particularly around 2020, indicate volatility clustering and periods of changing market conditions. This non-constant volatility is an important characteristic of financial returns and provides motivation for comparing a flexible LSTM model with the simpler Linear Regression approach.

3.4 Stationarity, autocorrelation and volatility clustering

The ADF statistic for returns is -15.6614 with $p < 0.001$, while the KPSS statistic is 0.09684 with a reported $p = 0.10$. Read together, these tests provide strong evidence that the return series is stationary in its level over the sample, at least in the broad sense tested by these procedures.

The significant Ljung-Box statistics for raw returns indicate that the series is not an ideal independent-noise process. More strikingly, squared returns are strongly autocorrelated and the ARCH-LM test is highly significant. This is consistent with volatility clustering of the type modelled in the ARCH/GARCH literature (Engle 1982). The result has two consequences. First, a model may exploit volatility regimes even when next-day direction remains hard to predict. Second, a synthetic benchmark based on i.i.d. Gaussian returns cannot reproduce the dependence and tail behaviour of the real series, so its error scale is not a perfect apples-to-apples null.

3.5 Feature correlation

As part of the exploratory analysis, the correlation between the current return and the next-day target was also examined. The largest reported absolute correlation is approximately 0.068 for current return versus the next return. However, current return was included only as an exploratory diagnostic and was not used as one of the seven final model predictors. Lag-1, lag-2 and lag-5 correlations with the target are approximately -0.0059, -0.0217 and -0.0348 respectively. Volume change is essentially uncorrelated with the target at 0.0022. These values suggest that any learnable conditional-mean signal is likely to be small.

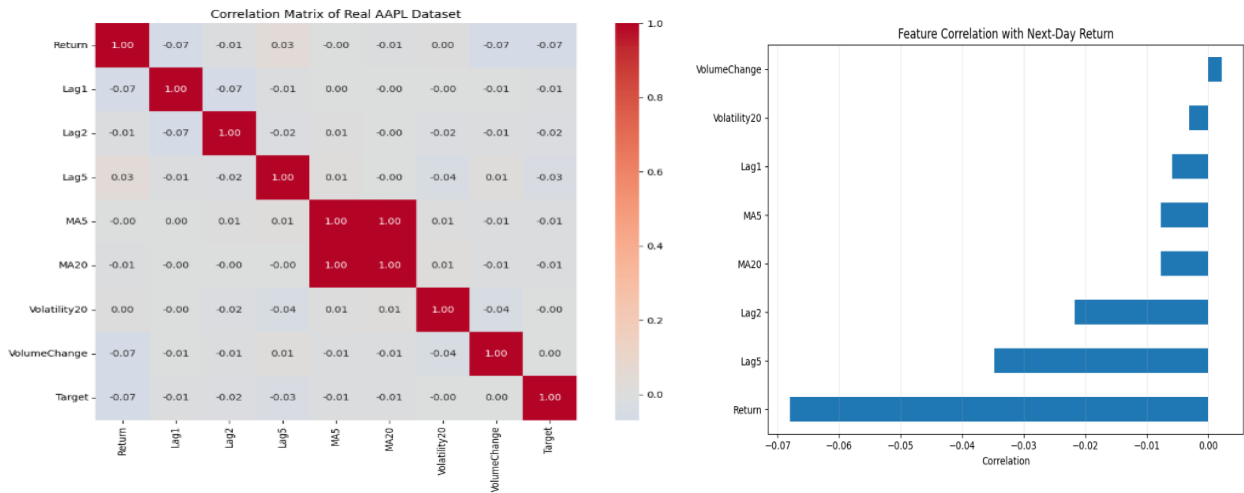


Figure 3.5: Feature Correlations with Next-Day AAPL Returns and Correlation Matrix of the Real AAPL.

The two price moving-average features are almost perfectly correlated ($r \approx 0.9985$). In a linear regression this collinearity can produce large, offsetting coefficients without providing additional information, and in a neural network it introduces redundant inputs. The fitted linear coefficients indeed have opposite signs for MA5 and MA20. A revised design could replace the pair with a scale-free spread such as $(MA5 - MA20)/P_t$, which is more interpretable and avoids feeding raw price levels with near-duplicate information.

3.6 Chronological split and scaling

The final real dataset is split chronologically into 80/20 split i.e. 1,994 observations from 4 February 2015 to 4 January 2023 form the training period, and 499 observations from 5 January 2023 to 30 December 2024 form the test period. MinMax scaling is fitted on training predictors and then applied to the test predictors. This avoids direct future-to-past leakage through scaler parameters.

A crucial descriptive fact is that the training period is more volatile than the test period. The linear-regression training RMSE is 0.018795 compared with test RMSE 0.013548, and the corresponding LSTM values are 0.018875 and 0.013304. This difference in target scale means that a negative test-minus-train error gap cannot, by itself, be interpreted as evidence against overfitting. The later discussion therefore distinguishes raw error gaps from scale-normalised generalisation evidence.

4. Methodology and Implementation

4.1 Research design

The study follows a controlled comparative forecasting design intended to separate model performance from evidence of genuine predictability. The analysis proceeds through four main stages. First, historical AAPL data are transformed into daily returns and a set of lagged, rolling and volume-based predictors. Second, Linear Regression and LSTM models are trained using a chronological division of the data, with a naive mean-return forecast providing a simple benchmark. Third, the same general modelling procedure is applied to a synthetic random-return series containing no intentionally introduced serial predictive mechanism. Finally, model performance is assessed using forecast errors, directional accuracy, residual diagnostics and statistical comparison.

The purpose of this design is not simply to identify the model with the lowest error. Instead, each stage addresses a different aspect of predictive validity: the naive benchmark establishes whether model complexity provides useful improvement; the chronological split tests performance on unseen observations; residual diagnostics examine whether systematic structure remains after forecasting; and the synthetic benchmark examines whether apparently successful modelling behaviour can also arise in a setting without an intentionally inserted predictive signal.

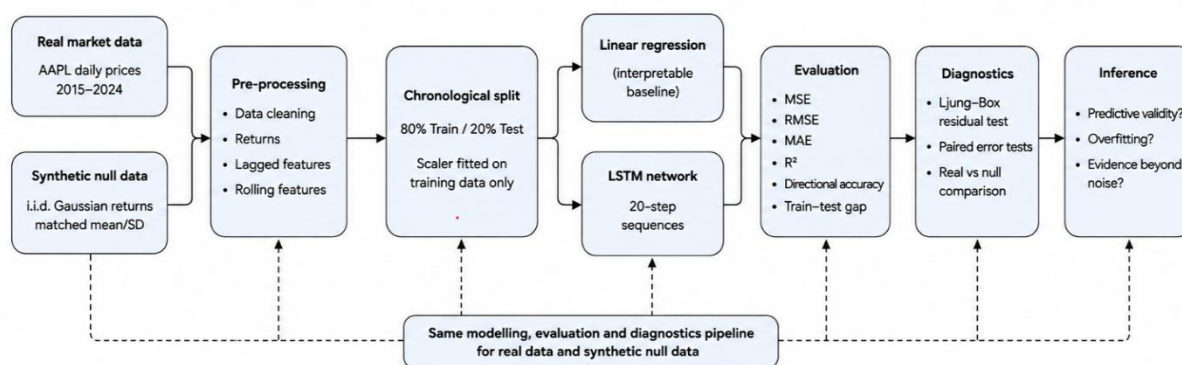


Figure 4.1: Updated research workflow. Real and synthetic data pass through comparable preparation, model fitting, out-of-sample evaluation and diagnostic stages before conclusions about validity and overfitting are drawn.

4.2 Naive mean-return benchmark

The naive model provides the mean return for each test observation for the period of training. Given that the loss function is the square error, the mean is a natural benchmark as the in-sample square error is minimized by it among all constant forecasts. This is one of the major characteristics of the directional forecast. Since the mean return of AAPL is positive, the naive forecast is positive each day, and hence its directional forecast equals the fraction of positive test returns. The performance of a complicated forecasting model must not be evaluated at 55% if a simple rule can achieve the same or better.

4.3 Linear regression

The linear model estimates are given by,

$$\hat{r}_{t+1} = \beta_0 + \sum_{j=1}^p \beta_j x_{j,t}$$

where $x_{j,t}$ are the scaled lag, rolling and volume features. Ordinary least squares provide an interpretable reference point and allow fitted coefficients to be inspected. The model is fitted once on the training set and evaluated without refitting on the held-out period.

The reported coefficients are small for lag and volume features, while the MA5 and MA20 coefficients are about +0.0273 and -0.0279. Given their near-perfect correlation, these coefficients should not be read as stable causal effects; they largely act as a contrast between two highly redundant price-level variables.

4.4 LSTM model

For the recurrent model, 20-step sequences were created. The intended design is to use recent multivariate history to predict the next return.



Figure 4.2: LSTM architecture implemented in the study.

The first LSTM layer has 64 units and returns a sequence; a 0.20 dropout layer follows. A second 32-unit LSTM produces a vector that is passed through another 0.20 dropout layer, a 16-unit ReLU dense layer and a one-unit output. Training uses Adam, MSE loss, a maximum of 100 epochs, batch size 32 and early stopping with patience 10 on validation loss. The executed real-data run stopped after 19 epochs, while the synthetic-data run stopped after 21. A fixed random seed of 42 is set to improve repeatability. Early stopping is an appropriate regularisation device because it prevents continued optimisation after validation loss stops improving. However, the executed validation split = 0.2 is applied after constructing heavily overlapping sequences. Adjacent sequences share 19 of 20-time steps, so samples on opposite sides of the training-validation boundary may contain nearly identical raw observations. A stronger design would split the raw time series before sequence construction and include a small purge gap at the boundary.

A target-alignment issue was identified during the implementation audit. The target variable had already been shifted by one trading day before the 20-step sequences were constructed. The sequence-generation procedure then selected the target at index $i+20$. Consequently, the executed LSTM alignment contains an additional offset relative to the intended one-day-ahead forecasting definition. The LSTM results are therefore interpreted as an implementation-specific experiment rather than as a perfectly aligned one-day-ahead forecast. A corrected implementation should construct the sequence and target explicitly so that the final observation in the input window maps directly to the following trading day's return.

4.5 Synthetic random-data benchmark

The synthetic random-data generates i.i.d. Gaussian returns using the real series observed mean and standard deviation, then constructs lagged returns, rolling return statistics, volatility and a random volume-change feature. The purpose is conceptual: if a model produces apparently strong predictive performance on data generated without serial signal, the evaluation procedure may be detecting sample artefacts. This is a useful first null, but it is deliberately simple. The real data exhibit heavy tails, return autocorrelation and pronounced ARCH effects, none of which are reproduced by an i.i.d. Gaussian generator. Moreover, the real-data MA5 and MA20 features are rolling means of closing price, whereas the synthetic benchmark constructs those features from returns. The feature spaces are therefore not strictly equivalent. The random benchmark is consequently treated as a diagnostic control rather than a formal hypothesis test that real data are “more predictable” whenever its MSE is larger.

4.6 Evaluation metrics

The primary loss is mean squared error,

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

RMSE is its square root and returns the error to the scale of returns

$$RMSE = \sqrt{MSE}$$

MAE is

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

which is less dominated by extreme errors.

The coefficient of determination is calculated in the usual form

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

A negative test R2 means that the model’s squared error is worse than predicting the relevant sample mean under the convention used by the evaluation function.

This metric is economically intuitive but insensitive to magnitude. It is therefore interpreted alongside, not instead of, continuous losses. The study defines an overfitting gap as test MSE minus training MSE. A large positive gap is consistent with overfitting, while a negative gap means the test period has lower raw MSE. Because financial volatility changes through time, this diagnostic is descriptive rather than definitive. A better version would normalise loss by a period-specific naive benchmark or evaluate repeated rolling windows.

4.7 Residual and statistical diagnostics

Ljung-Box tests are applied to model residuals at lags 5, 10 and 20. Large p-values indicate that remaining residual autocorrelation is not statistically detected at those lags. This is useful but not sufficient for forecast validity: white residuals can coexist with forecasts that are no better than a simple baseline. The study also performs paired comparisons of linear-regression and LSTM absolute errors over 479 aligned sequence observations. A paired t-test assesses the mean loss difference under parametric assumptions, and a Wilcoxon signed-rank test provides a rank-based alternative. The standardised paired effect size is also reported. Since the data are time-ordered, these tests are best interpreted as exploratory; a Diebold-Mariano procedure with a suitable long-run variance estimate would better address serial dependence in loss differentials.

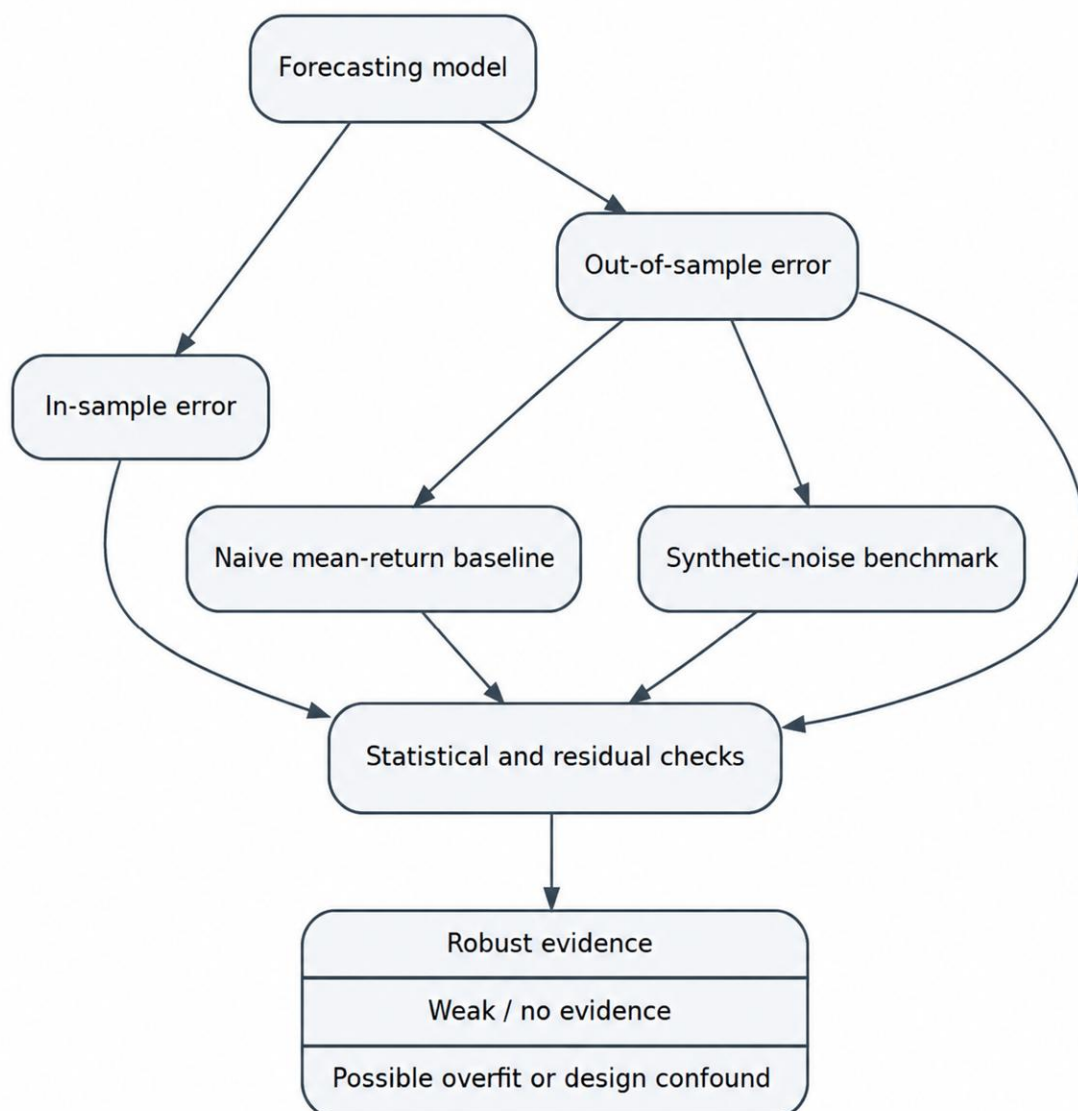


Figure 4.3: Interpretive validation logic. No single statistic is treated as proof of predictive validity, evidence is assembled from hold-out loss, simple benchmarks, a synthetic control and statistical diagnostics.

4.8 Evaluation strategy and validity checks

Table 4.1: Evaluation and validity checklist

Validity check	Purpose	Implementation
Chronological train-test split	Prevent future observations from being used to train the model	80% training and 20% testing in temporal order
Training-only scaling	Prevent information from the test period entering the fitted transformation	MinMax scaling fitted on training data only
Naive benchmark	Determine whether model complexity provides useful improvement over a simple forecast	Mean-return benchmark evaluated on the same test period
Multiple forecast metrics	Avoid judging models using a single measure of performance	MSE, RMSE, MAE, (R^2) and directional accuracy
Train-test comparison	Identify the conventional pattern of overfitting	Training and test errors compared for each model
Residual diagnostics	Examine whether systematic temporal structure remains after modelling	Ljung–Box tests applied to model residuals
Statistical comparison	Determine whether differences between models are larger than expected from sampling variation	Paired (t)-test and Wilcoxon signed-rank test for LR versus LSTM errors
Synthetic random benchmark	Examine whether apparent predictive performance can arise without an intentionally inserted predictive mechanism	Linear Regression and LSTM applied to synthetic random returns
Directional benchmark comparison	Assess whether models improve on a simple directional rule	Model directional accuracy compared with the naive benchmark
Reproducibility checks	Reduce dependence on undocumented implementation choices	Fixed random seed and recorded model architecture/hyperparameters

The evaluation strategy was designed to avoid relying on a single measure of model success. Forecast accuracy is assessed using MSE, RMSE, MAE and R2, while directional accuracy provides a complementary measure of whether the predicted sign of the return is correct. These metrics address different aspects of forecasting performance and reduce dependence on a single measure of model quality.

The naive mean-return benchmark provides a minimum-performance reference. A model that performs only marginally better than this benchmark provides weaker evidence of useful predictive information than a model that produces a substantial and statistically reliable improvement. The models are evaluated using a chronological train-test split so that observations from the future are not used to train the models. Feature scaling is fitted using the training data before being applied to the test data, reducing the risk of information leakage.

The comparison between training and test performance is used as a diagnostic for conventional overfitting. However, because financial volatility changes over time, the train-test gap is not interpreted in isolation. Residual diagnostics are therefore used to examine whether systematic temporal structure remains after forecasting. Statistical tests provide an additional assessment of whether differences between model errors are sufficiently large to distinguish them from sampling variation.

The synthetic random-data experiment provides an additional validity check. The same general modelling framework is applied to a return series in which no serial predictive mechanism is intentionally introduced. If the modelling procedure produced apparently strong forecasting performance under this setting, confidence in similar performance on the real data would be reduced. The synthetic experiment is therefore treated as a diagnostic control rather than as a formal statistical proof that the real-data results are genuine.

4.9 Implementation environment

The analysis was implemented in Python using a Jupyter Notebook hosted on the Kaggle platform. The computational environment used Python 3.12.13. The main libraries included NumPy and pandas for data manipulation, SciPy and stats models for statistical analysis, scikit-learn for Linear Regression, preprocessing and evaluation, TensorFlow/Keras for the LSTM model, Matplotlib and seaborn for visualisation, and yfinance for obtaining the AAPL market data. The experiments were conducted using the Kaggle CPU environment, with no GPU required for the analysis. A random seed of 42 was used where applicable to improve the reproducibility of the numerical and neural-network experiments.

For transparency and reproducibility, the complete implementation of the analysis is available in the accompanying Kaggle notebook: [Stock market forecasting](#)

5. Results

5.1 Overall out-of-sample performance

Model	MSE	RMSE	MAE	R^2	DA
Naive mean return	0.000181	0.013460	0.010005	-0.001710	56.31%
Real linear regression	0.000184	0.013548	0.010102	-0.014770	49.90%
Real LSTM	0.000177	0.013304	0.009872	-0.000139	55.32%
Random linear regression	0.000296	0.017205	0.013821	-0.007362	51.52%
Random LSTM	0.000297	0.017225	0.013861	-0.005902	52.84%

Table 5.1: Final out-of-sample model comparison

The LSTM has the lowest test MSE, RMSE and MAE among the executed models. Relative to linear regression, its MSE is approximately 3.8% lower; relative to the naive mean-return baseline, the reduction is approximately 2.2%. The numerical improvement is therefore real in the reported table but small. The negative or near-zero R^2 values indicate that none of the fitted models explains a meaningful share of out-of-sample variance under this metric.

The directional result points in a different direction. The naive constant-positive forecast is correct on 56.31% of test days, compared with 55.32% for the LSTM and 49.90% for linear regression. The LSTM therefore does not outperform the naive benchmark on sign prediction. This is an important warning against choosing a single preferred metric after seeing the results.

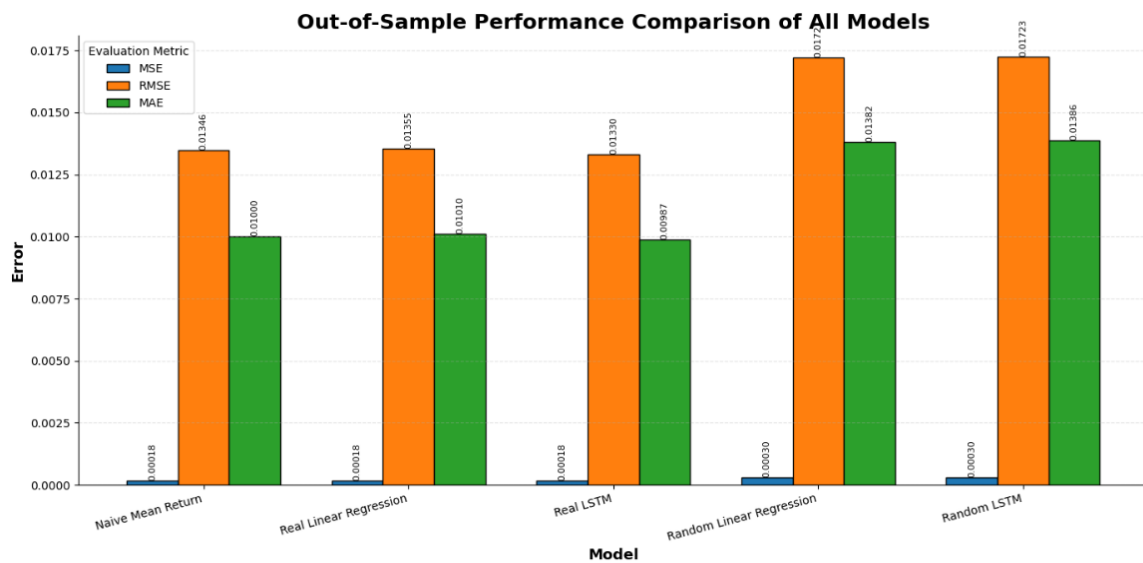


Figure 5.1: Out-of-Sample Performance Comparison of Real and Simulated Random Data

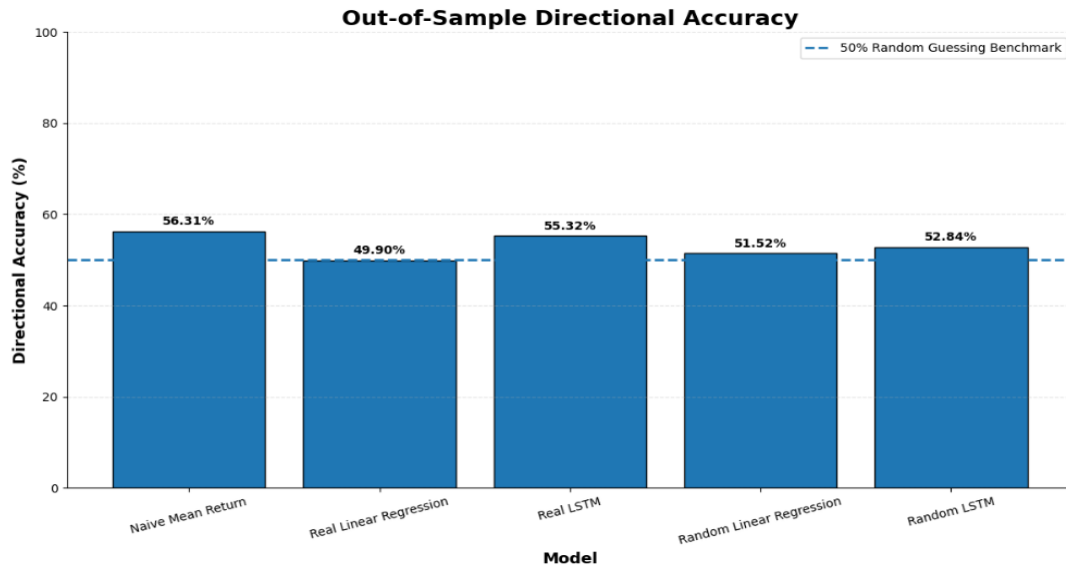


Figure 5.2: Out-of-sample directional accuracy. The naive positive-return baseline is the best of the real-data models on this metric.

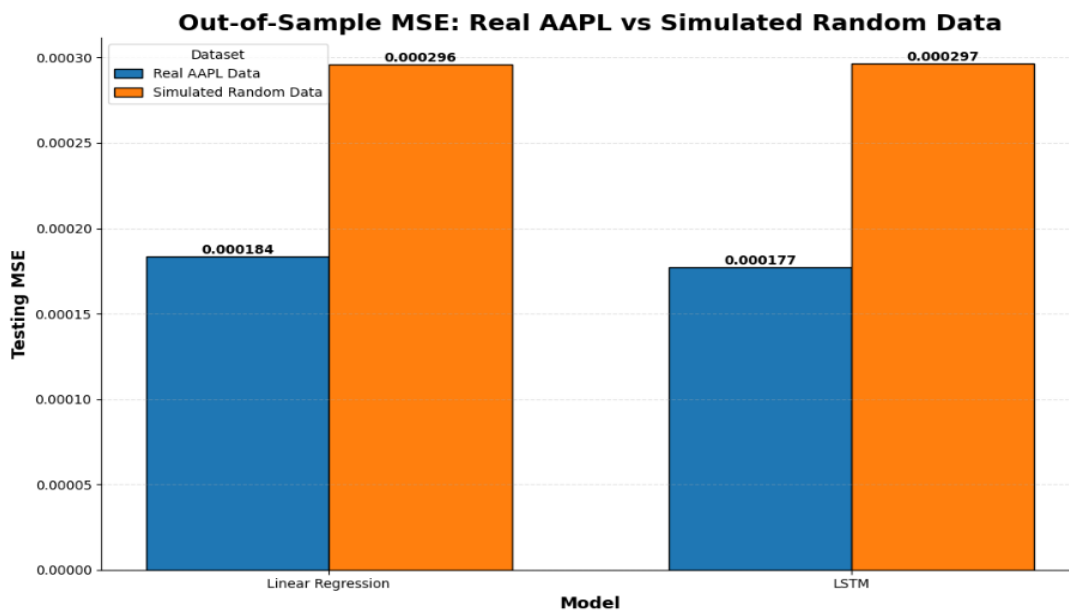


Figure 5.3: Out-of-Sample MSE Comparison: Real AAPL vs Simulated Random Data

5.2 In-sample versus out-of-sample performance

Table 5.2 presents the performance of the fitted models on the training and test sets. The test minus training difference in MSE is negative for all models: -0.000169 for the real data linear regression model, -0.000179 for the real data LSTM, and -0.000016 for both synthetic data models. Therefore, the data is inconsistent with the standard picture where training error is much lower than the test error. But negative gaps between MSE cannot be considered as evidence of lack of overfitting. First of all, training and test intervals in real data have different features of returns and volatilities. The training interval has more periods with greater variations and larger absolute errors compared to the test interval, and this leads to greater MSE despite the absence of overfitting.

Model	Train MSE	Train RMSE	Test MSE	Test RMSE
Real linear regression	0.000353	0.018795	0.000184	0.013548
Real LSTM	0.000356	0.018875	0.000177	0.013304
Random linear regression	0.000312	0.017667	0.000296	0.017205
Random LSTM	0.000313	0.017687	0.000297	0.017225

Table 5.2: Training and test performance for fitted models.

The comparison with the random benchmark offers yet another form of confirmation. In the case of the synthetic-data models, there is very little difference in terms of training and testing errors. On the other hand, both real-data models generate much lower test MSE compared to the models generated by the synthetic benchmark data. This implies that there is something captured in the real-data models that is not present in the synthetic benchmark model; however, by itself it does not confirm the existence of economic predictability.

The assessment of overfitting involves consideration of several pieces of evidence. The dissertation includes consideration of out-of-sample performance, comparison with the naive benchmark, directional prediction success, residuals diagnostics, and statistical tests of differences in forecast errors. This approach becomes important as a single split into training and testing samples does not confirm stability of predictions across all market conditions.

5.3 Linear-regression coefficients and residuals

Feature	Coefficient
Lag 1 return	-0.002523
Lag 2 return	-0.007224
Lag 5 return	-0.012228
MA5	0.027333
MA20	-0.027897
20-day volatility	0.000034
Volume change	0.000781

Table 5.3: Linear-regression coefficients for the real-data model.

Table 5.3 gives the values of the coefficients in the estimated Linear Regression model based on the real data set. The coefficients of Lag1, Lag2, and Lag5 are negative and rather small in value, which implies weak linear relationship between the past returns and the return on the next day. Also, the coefficients of Volatility20 and VolumeChange have rather small values close to zero and thus imply weak linear contribution of the corresponding variable into the forecasted return.

As far as the two moving-average variables are concerned, they have the largest coefficients in absolute value: $MA5 = 0.027333$ and $MA20 = -0.027897$. These values cannot be considered independently since there is very high correlation between the two variables, namely their correlation equals approximately 0.9985. Thus, there is much multicollinearity, and the values of the coefficients might be unstable, which makes it impossible to interpret their opposite signs as evidence of the distinct short- and medium-term effect.

Further insight into the existence of unexplained temporal patterns can be gained from the residual diagnostics. For the Linear Regression, the Ljung–Box test reports p-values of 0.140, 0.379 and 0.419 at lags 5, 10 and 20, correspondingly. In the case of LSTM, p-values for the same lags are 0.159, 0.470 and 0.723, correspondingly. As all these values are larger than the conventional significance level of 5%, we fail to reject the null hypothesis that there is no residual autocorrelation at these lags. Therefore, none of the models left statistically significant autocorrelation in its residuals.

In contrast to the detected autocorrelation in the return series in the exploratory part, the fact that both models removed it somewhat indicates that they were able to capture some of the short-memory properties of the returns. Nonetheless, the lack of any residual autocorrelation does not prove that the models are providing forecasts of value. Specifically, out-of-sample R^2 of the Linear Regression fitted on real data is roughly equal to -0.0148 , which means that its test predictions are worse than a benchmark forecast that uses the mean return.

On balance, the findings from the coefficients and residuals show that the models contain little short-term statistical structure, but there is no strong indication of predictability over one day ahead. This difference is crucial for the dissertation since the removal of residual autocorrelation does not amount to an indication of prediction. Thus, the findings have to be combined with error measures on out-of-sample data, the naive predictor, directional accuracy, random data, and prediction errors comparisons.

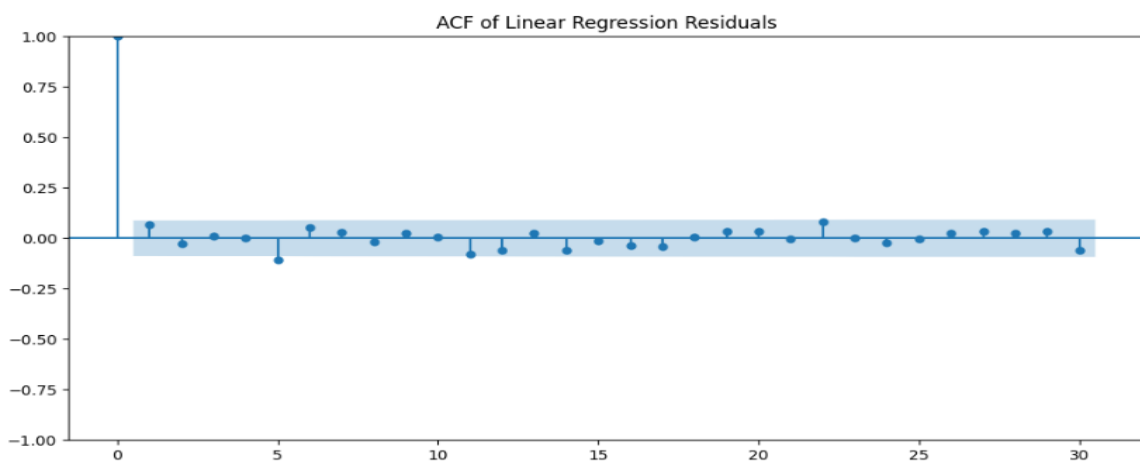


Figure 5.4: Autocorrelation Function of Linear Regression Residuals

5.4 LSTM learning behaviour

The real data LSTM network stopped at 19 iterations due to early stopping criteria, implying that the validation loss had stopped improving sufficiently during training. The training MSE for the LSTM was 0.000356, slightly higher than the training MSE of the linear regression model, which is 0.000353. This implies that the LSTM had not achieved a much better fit to the training data.

In the case of the test data, the LSTM had a slightly better MSE (0.000177) compared to linear regression (0.000184), implying a slight improvement in numerical forecasting accuracy. However, the difference is small and does not provide strong evidence of genuine forecastability.

Consequently, although there is a small improvement from the LSTM compared to the linear model, the increase in the model complexity does not give an obvious economic forecasting advantage. This reinforces the main thesis of the dissertation, where we see that prediction error alone does not constitute sufficient evidence of genuine predictability, especially since performance should also be gauged against the naive and random data benchmarks.

5.5 Real versus synthetic data

Both the synthetic models yield test MSEs of approximately 0.000296–0.000297, much higher than those observed in the real data models that achieve test MSE of approximately 0.000177–0.000184. Additionally, both the synthetic models yield negative R^2 values, while their directional accuracy is similar to chance at 51.52% for linear regression and 52.84% for LSTM. On the synthetic data set, the two models exhibit near identical performances, which can be attributed to the absence of any serially predictive structures.

At first glance, the smaller error yielded on the real AAPL returns could indicate the presence of true predictability. However, such a result should be taken with caution since the synthetic benchmark fails to recreate all the features of the real returns series, including the dynamics of volatility, tails, and the exact feature relationships. The fact that MSE is dependent on the variance of the target makes part of the explanation for the observed difference in error.

In general, the experiment on synthetically generated data supports but does not prove anything. In other words, it is demonstrated that the application of neither linear regression nor the LSTM model leads to high predictive power when working with data which was synthetically created without using any artificial serial signal. Nevertheless, this does not mean at all that the better results obtained with respect to AAPL are reliable and could be used for economic profit.

5.6 Paired statistical comparison

This study compares the 479 paired absolute forecast errors of the Linear Regression and LSTM models. In the case of the paired t-test, the obtained statistic value is 1.9338, p-value = 0.0537, whereas the statistic value for the Wilcoxon signed-rank test is 52,242 and the p-value equals to 0.0840. The estimated standardised paired effect size is 0.0884 and it means that the difference between the two models is relatively small.

At the standard 5% significance level, neither test can reject the null hypothesis of the equality of forecast losses. Hence, even though the LSTM model provides the minimum observed test

MSE value, it does not allow to claim that the difference between the two models is statistically significant. It also indicates that the practical difference between the two models is relatively small, due to the small effect size.

It should be noted that these tests compare only the LSTM and the Linear Regression models. In other words, they check whether the difference in the forecast losses of the two models is statistically significant. However, they do not compare these models with the naive mean-return forecast. Hence, the estimated improvement in MSE by about 2.2% relative to the naive model cannot be considered statistically significant.

6. Discussion

6.1 RQ1: How predictable are next-day AAPL returns?

Predictive skill of next-day returns is not high according to the results. While LSTM shows the minimum continuous errors in comparison with other fitted models, the improvement of LSTM compared to Linear Regression is insignificant and its R^2 is still close to zero. Moreover, the directional accuracy of LSTM is no better than the constant-positive benchmark model. Linear Regression does not perform well in both forecasting error and directional accuracy compared to the constant-positive benchmark model. In general, there is little evidence of any significant next-day predictability.

Such a result is in agreement with the definition of weak-form market efficiency when information from prices and returns should have little or no predictive power. Despite that serial dependence was found using EDA, it is shown that statistical dependence does not imply predictability. Very low correlations of historical variables with next-day returns and negative or zero R^2 also justify this claim.

The results further indicate that increased model complexity does not necessarily translate to better prediction accuracy. Even though the LSTM marginally improves the prediction error compared to Linear Regression, the statistical test does not find any convincing difference between the two models. The above implies that the increased complexity of the LSTM does not add any useful value in predicting returns in this situation.

Overall, the evidence provides only limited evidence of a stable next-day predictive signal in AAPL returns using the available historical features. The performance obtained is not sufficient to show any predictive signal.

6.2 RQ2: What does the in-sample/out-of-sample gap say about overfitting?

The negative train-test MSE gaps do not provide evidence of the conventional pattern of overfitting, but they cannot be interpreted as evidence that overfitting is absent. The training and test periods differ substantially in volatility, making raw MSE comparisons across periods difficult to interpret. The training period includes higher-volatility episodes, including the market conditions around 2020, whereas the 2023–2024 test period is relatively calmer. Because MSE and RMSE are sensitive to the scale of the target, lower test errors may partly reflect differences in market volatility rather than stronger model generalisation.

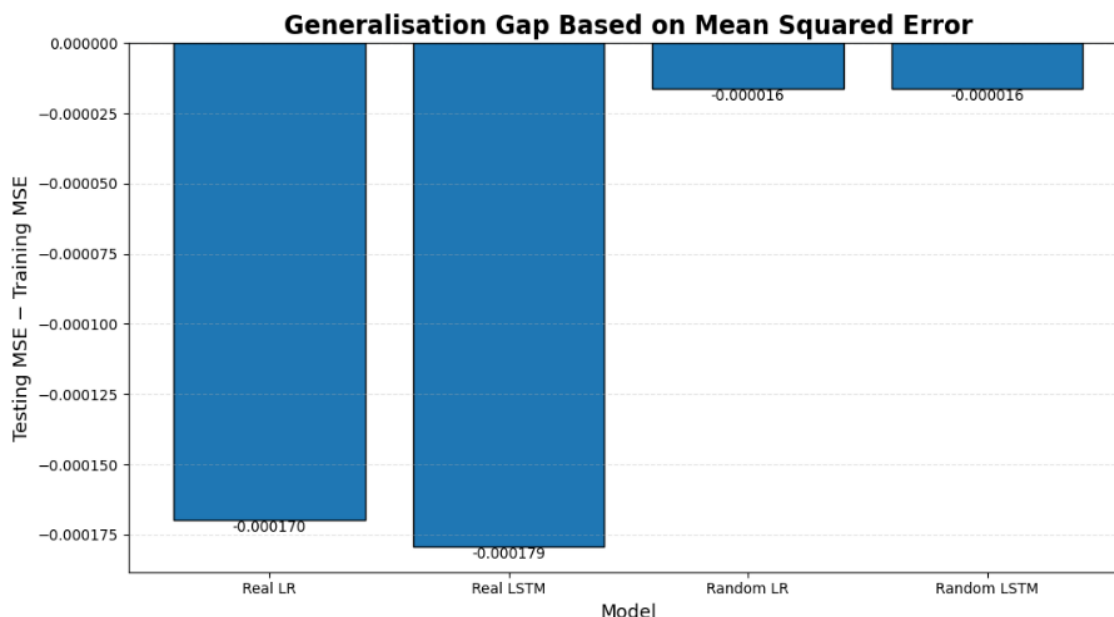


Figure 6.1: Train–Test MSE Generalisation Gap

These findings do not offer strong evidence that the models avoid simple memorisation, but they do not provide evidence that the models can generalise well. The most appropriate way to measure generalisation would be to compare the model errors with a naive approach in the same period. The LSTM also does not give good reasons to believe that an increase in model complexity leads to better generalization. Even though the model performs better in terms of lower test MSE compared to Linear Regression, the difference between the two models is marginal and the paired statistical tests do not support the existence of a reliable improvement. In addition, problems with sequence-target alignment and design of validation process were revealed during the model audit, thus weakening the conclusions about the generalization properties of LSTM.

Thus, the train/test gap in this case is not an indication of conventional overfitting, however, it is not enough to exclude other forms of overfitting. The reason for that is the fact that backtest overfitting may appear due to the repeated process of model selection and dependence on a specific sample of data even without the test error being higher than the training one.

6.3 RQ3: Can real data performance be differentiated from random patterns?

The synthetic experiment yields some indications that the models are not able to build up any meaningful predictive performance from random patterns. Both random patterns' models have negative R^2 and directional accuracy close to randomness, and moving from Linear Regression to LSTM model does not provide much progress. This is congruent with the fact that there is no intentionally built serial predictive pattern. However, the lower forecasting error in the case of real AAPL data cannot be considered conclusive evidence of the existence of any predictable pattern. The synthetic benchmark fails to replicate the properties of financial returns, such as heavy tails and clustering of volatility. Hence, MSE differences can be partially driven by distributional and variance differences between the two sets of data.

The above random data experiment can be considered as supportive but not definitive evidence for research question RQ3. The above experiment proves that the modeling pipeline cannot automatically generate a high value of R^2 or directional performance when used on i.i.d random returns. However, it cannot confirm the probability of the actual randomness of the above AAPL performance.

In the future, a more definitive test should be carried out by using more realistic nulls such as the block permutations, stationary bootstrap or the GARCH simulations, among others. This experiment should be replicated many times to come up with a null distribution that will be used to compare with the above AAPL performance.

6.4 Statistical versus economic significance

Statistical significance does not necessarily imply economic relevance. Even though the model decreases forecasting error, it does not guarantee that it generates trading profits for a model. The current research does not consider transaction costs, bid-ask spread, market impact, turnover, and risk-adjusted returns. Hence, one cannot interpret it as a profitability analysis.

Moreover, one needs to consider directional accuracy with respect to the benchmark. For instance, a naive positive return forecast yields 56.31% directional accuracy. Hence, if the model is able to generate a similar value, it will not imply any additional information. Thus, there needs to be a significant improvement in order for the model to surpass the benchmark and not just beat it in terms of directional accuracy.

Hence, this analysis represents statistical forecasting performance rather than an economically relevant result. Future research can incorporate transaction costs and develop trading rules that allow calculating net returns, Sharpe ratio, maximum drawdown, and turnover.

6.5 Implication to the Efficient Market Hypothesis

This result is somewhat compatible with weak-form market efficiency, but it is not a test for EMH. LSTM performs slightly better than linear regression in terms of MSE, which implies that there is a slight presence of weak conditional structure. The fact that the model did not yield any positive out-of-sample R^2 , the superior performance of the naive benchmark in terms of directionality, and the non-significance of the difference between linear regression and LSTM, however, suggests that there is no predictive edge that can be used by investors.

EDA makes a valuable contribution in clarifying that dependence does not necessarily translate into return predictability. Even though Ljung–Box tests have indicated presence of some serial dependence in returns and ARCH test strongly suggests presence of volatility clustering, it does not imply predictability of the excess returns. Volatility can be predicted even if the conditional mean of returns is hard to predict.

Overall, the results suggest that detectable time-series structure does not automatically translate into useful next-day return prediction. This is consistent with the weak-form EMH interpretation that historical price information may contain statistical patterns without providing a sufficiently stable opportunity to generate abnormal returns.

6.6 Limitations

There are a number of limitations associated with the study that affect its ability to have wide applicability. First, the research is conducted based on a single asset, and therefore its results cannot automatically be applied to any other asset or any other company. Second, the model evaluation process is carried out using only one chronological train-test split 80/20 and not by using rolling or expanding windows, which means that the results can be dependent on the regime of the market periods. Third, the set of features is fairly limited and includes highly correlated moving average variables. This could mean that the models will not be able to utilize other sources of predictive signals. Fourth, there is an alignment problem with the forecasting horizon of the constructed LSTM sequence that does not completely match the one-day-ahead forecasting objective of the research.

Fifth, the random benchmarking procedure fails to replicate key properties of actual stock market returns like fat-tails and volatility clustering. Thus, discrepancies between real and artificial forecasting errors do not amount to a proper statistical analysis for proving actual predictability. Sixth, the statistical comparison concerns only Linear Regression vs. LSTM, without applying any forecasting test such as Diebold-Mariano. Seventh, there were no package versions and data download parameters completely set, thus exact replicability might be an issue. Lastly, the work is assessing forecasting performance statistically, but not economically. There is no consideration of trading costs, spreads, portfolio limits and risk-adjusted profits. The results should thus be understood as evidence on predictive ability, and not profitable trading strategy.

Conclusions and Future Work

7.1 Conclusions

In this thesis, I studied whether Linear Regression and LSTM can predict the next-day returns of AAPL while providing overfitting-robust evidence for their ability to do so. In addition to the out-of-sample approach, I considered the naive model benchmark, residual analysis, artificial random-data control and statistical comparison of model errors.

The overall finding is that there is no compelling evidence of any predictive power on next-day returns in the information used for predictions. The LSTM gave the smallest forecast error, yet the improvement over other models was modest and did not imply superiority of the model in any way. Particularly, the model did not outperform the naive model regarding direction prediction. The study also illustrates the limitations of the common interpretation of the train-test gap in financial forecasting. The test period was less volatile than most of the training period; hence, smaller forecast errors in the test period could not be regarded as evidence of good generalisation of the models. More generally, the lack of a common overfitting problem does not mean that model selection is not biased by the specificities of the historical sample.

However, when the same modelling technique was used on data that did not have any deliberately designed predictive signal in the process of their generation, neither Linear Regression nor LSTM yielded good prediction results. This confirms the value of a no-signal benchmark as a useful test, though it should be pointed out that the simulated data do not exhibit the full range of properties of real-life stock return series.

The main lesson from the study is therefore methodological rather than the identification of a superior forecasting model. In financial prediction, a model that produces the smallest error is not necessarily a model that has discovered a genuine and reusable signal. A stronger claim of predictability requires performance that survives comparison with simple benchmarks, unseen data, appropriate null experiments and statistical scrutiny.

Within the scope of the present experiment, the evidence is therefore consistent with a cautious interpretation of weak-form market efficiency. The study does not demonstrate that AAPL returns are completely unpredictable; rather, it shows that the available historical features and implemented models do not provide sufficiently strong evidence to distinguish a useful predictive signal from noise with confidence.

7.2 Future work

There are a number of improvements that can be made to enhance future studies on this topic. For instance, the experiment can adopt a rolling window approach for validating the models, rather than having an 80/20 split at a point in time. Another improvement would be to conduct the analysis for other stocks and assets and not just for Apple.

Finally, the artificial benchmark can be enhanced further. While the use of Gaussian distributed independent random variables is a good start, future research should consider the use of more sophisticated techniques such as block permutation, stationary bootstrap and GARCH simulation for generating the benchmark. Such a benchmark will be able to maintain features such as volatility clustering and fat tailed distributions.

LSTM should also be refined for an exact target matching the desired one-day-ahead forecasting period and the avoidance of validation leakage. Rolling validation and prediction-specific statistical test such as Diebold-Mariano test would make for a stronger comparison of models' predictive power. Further, other methods can also be applied, such as the initially proposed Random Forest, which allows for comparing models' complexities.

Lastly, economic performance can also be considered in further studies. Predictions can be transformed into a certain predefined trading strategy and its performance can be judged by such criteria as transaction cost, turnover, Sharpe ratio, and maximum drawdown. The LSTM shows a small observed numerical improvement over Linear Regression, but the paired statistical tests do not provide sufficient evidence that this difference is statistically significant at the 5% level.

Overall, the most important direction is to move from a single-split exploratory experiment toward repeated, temporally robust and economically evaluated testing. This would provide stronger evidence for distinguishing genuine predictive information from patterns that arise through sampling variation, model selection or backtest overfitting.

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